

Luca Morreale

 <https://luca-morreale.github.io/>

 luca.morreale93@gmail.com

 LinkedIn

 Google scholar

 Github

Research Scientist in Generative AI, Multimodal Systems, and 3D Vision. Track record of translating frontier research into practical AI systems with real-world impact. Experience across Samsung, Meta, Adobe, and UCL.

Education

- 2019 – 2024  **Ph.D., University College London** Computer science
Thesis title: *Neural Surfaces*
Advisor: Niloy J. Mitra.
Doctoral research on machine-learning methods for shape matching, developing approaches for geometric correspondence across complex 3D data. Published research advancing the intersection of geometric processing and machine learning.
- 2015 – 2017  **M.Sc., Politecnico di Milano** in Computer Engineering.
Computer Vision and Robotics perception
- 2012 – 2015  **B.Sc., Politecnico di Milano** in Computer Engineering.
Software engineering

Experience

- 2024 -  **Samsung Research**, Full time employee
Led development of world models and diffusion-based image/video systems focused on controllability, robustness, and efficient post-training. Owned research directions, aligning exploration with product.
- 2023  **Meta Reality Labs**, Research intern
Led research on language-driven scene geometry manipulation, enabling natural-language editing of 3D environments through multimodal understanding and controllable geometric transformations.
- 2021  **Adobe Research**, Research intern
Developed novel methods for shape compression, deformation, and editing, improving efficiency and fidelity for large-scale 3D asset creation pipelines. Contributed research prototypes for next-generation creative tooling focused on intelligent geometry processing.
- 2019  **Università della Svizzera Italiana**, Research Assistant
Led research in geometric deep learning for point clouds, designing neural architectures for representation learning, segmentation, and recognition on irregular 3D data.
- 2018 – 2019  **Politecnico di Milano**, Research Assistant
Led research on real-time 3D reconstruction and semantic simplification, transforming noisy spatial information into usable scene representations.

Research Publications

- **NanoFLUX: Distillation-Driven Compression of Large Text-to-Image Generation Models for Mobile Devices**
R. Chavhan, M. Chadwick, A. G. C. P. Ramos, **L. Morreale**, M. Noroozi, and A. Mehrotra
International Conference on Machine Learning (ICML) 2026.
- **FraQAT: Quantization Aware Training with Fractional bits**
L. Morreale, A. G. C. Ramos, M. Chadwick, M. Noroozi, R. Chavhan, A. Mehrotra, and S. Bhattacharya
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings (CVPRF) 2026.

- **RFDM: Residual Flow Diffusion Model for Efficient Causal Video Editing**
M. Salehi, M. Noroozi, **L. Morreale**, R. Chavhan, M. Chadwick, A. G. Ramos, and A. Mehrotra
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026.
- **Edit: Efficient diffusion transformers with linear compressed attention**
P. Becker, A. Mehrotra, R. Chavhan, M. Chadwick, **L. Morreale**, M. Noroozi, A. Gil CP Ramos, and S. Bhattacharya
Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) 2025.
- **Upcycling Text-to-Image Diffusion Models for Multi-Task Capabilities**
R. Chavhan, A. Mehrotra, M. Chadwick, A. G. C. P. Ramos, **L. Morreale**, M. Noroozi, and S. Bhattacharya
International Conference on Machine Learning (ICML) 2025.
- **Guidance Free Image Editing via Explicit Conditioning**
M. Noroozi, A. G. Ramos, **L. Morreale**, R. Chavhan, M. Chadwick, A. Mehrotra, and S. Bhattacharya
arXiv preprint arXiv:2503.17593 2025.
- **Neural Semantic Surface Maps**
L. Morreale, N. Aigerman, V. G. Kim, and N. J. Mitra
Computer Graphics Forum Wiley Online Library 2024.
- **Neural convolutional surfaces**
L. Morreale, N. Aigerman, P. Guerrero, V. G. Kim, and N. J. Mitra
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2022.
- **Neural surface maps**
L. Morreale, N. Aigerman, V. G. Kim, and N. J. Mitra
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2021.
- **Dense 3D visual mapping via semantic simplification**
L. Morreale, A. Romanoni, and M. Matteucci
International Conference on Robotics and Automation (ICRA) IEEE 2019.
- **Predicting the next best view for 3d mesh refinement**
L. Morreale, A. Romanoni, and M. Matteucci
Intelligent Autonomous Systems 15: Proceedings of the 15th International Conference IAS-15 Springer 2019.
- **Learning and mining player motion profiles in physically interactive robogames**
E. L. Oliveira, D. Orrù, **L. Morreale**, T. P. Nascimento, and A. Bonarini
Future Internet MDPI 2018.

Patents

- **Neural network based 3D object surface mapping**
V. Kim, N. Aigerman, N. J. Mitra, and **L. Morreale** US Patent 11,869,132 Jan. 2024.

Awards

- 2022  **Rabin Ezra scholarship**
The Rabin Ezra Scholarship Trust has been set up to make available bursaries for post-graduate students specializing in computer graphics.